[INCH-POUND]
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May 7, 1997

COMMERCIAL ITEM DESCRIPTION

WATTMETER
(AUDIO POWER METER)

The General Services Administration has authorized the use of this Commercial Item Description, for all federal agencies.

1. SCOPE

1 Scope. This Commercial Item Description (CID) describes a wattmeter (audio power meter). This CID is meant as a minimum requirement for the wattmeter in which only those manufacturers which meet or exceed the following requirements are qualified per this CID. This wattmeter is used to measure the output power of audio devices in the frequency range of 20 Hz to 20 kHz.

2. SALIENT CHARACTERISTICS

2.1 Equipment covered by this CID shall be portable COTS equipment.and may be modified to the extent necessary to meet the following description.

2.2 Safety and Environmental. The equipment shall meet all safety and environmental requirements as specified in Class 3 of MIL-PRF-28800F.

2.3 Electrical Power Sources. The equipment shall operate from nominal commercial, military, and shipboard power sources of 115 VAC (+/-10%) at line frequencies of 50 Hz and 60 Hz. Power consumption shall not exceed 25 watts.

2.4 Temperature Requirements.

2.4.1 Operating Temperature. The equipment shall meet its performance and accuracy requirements in an operating environment of 0 to +50 deg C.

2.4.2 Nonoperating Temperature. The equipment shall meet its performance and accuracy requirements after being in a nonoperating (storage) environment of -10 to +60 deg C.

2.5 Mean Time Between Failures (MTBF). The equipment MTBF shall be 2,000 hours. The lower test MTBF shall be 2,000 hours and the upper test MTBF shall be 4,000 hours.

2.6 Pre-conditioning Requirement. Each deliverable wattmeter shall be capable of being subjected to the environmental screening effects of burn-in and thermal cycling, with thermal cycling being accomplished in the sample testing phase.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Technology & Industrial Services Division, SA-ALC/TILDD, Bldg. 171, Post C-12, 485 Quentin Roosevelt Rd., Kelly AFB TX 78241-6425.

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2.7 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments. These adjustments shall be provided by variable value components which are adjustable by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules and shall be front panel adjustment or front panel automatic calibration in watts and dBm. The calibration interval shall be a period of one year or greater based on an operating time of 2000 hours.

2.8 Performance Characteristics.

2.8.1 Frequency Range. The minimum frequency range of the wattmeter shall be 20 Hz to 20 kHz.

2.8.2 Power Range. The minimum power range of the wattmeter shall be 300 microwatts to 20 watts.

2.8.2.1 Power Accuracy. The minimum power accuracy of the wattmeter shall be +/- 7.5% or less at full scale deflection (FSD).

2.8.3 Impedance. The impedance of the wattmeter shall be at least 10 to 600 ohms.

2.8.3.1 Impedance Accuracy. The impedance accuracy of the wattmeter shall be +/- 7% or less for 70 Hz to 5 kHz.

2.8.4 Interface. The wattmeter shall be front panel controlled.

2.8.4.1 Connectors. The wattmeter shall have three (3) binding posts on the front panel. Two of the binding posts shall be for inputs, and one binding post shall be for chassis ground.

2.8.4.2 Front Panel Display. The wattmeter shall have an analog meter or digital readout display on the front panel which displays power readings in linear mode -- milliwatts/watts (dependent on range selected) and in logarithmic mode --dBm.

2.8.4.3 Power Range Switch. The wattmeter shall have a switch on the front panel for the selection of an appropriate power range for unit under test power output measurement.

2.8.4.4 Impedance Selection Switch. The wattmeter shall have a switch on the front panel to select the impedance (load) that will be applied to the unit under test.

2.9 Weight. The wattmeter shall not weigh more than 20 lbs.

3. REGULATORY REQUIREMENTS

3.1 The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS

4.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

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5. **PACKAGING**

5.1 Preservation, packing, and marking shall be as specified in the contract or order.

6. **NOTES**

6.1 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met. If a product manufactured to metric dimensions exceeds the tolerances specified in the
inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

6.2 Known Acceptable Products. The following products are known to have previously met the requirements of this CID and need not resubmit bid samples or certification. All other manufacturers must meet the requirements of this CID and submit to bid testing and certification as required.

MANUFACTURER	CAGE	P/N	NSN
Quadtech	0PK96	1840-9701	6625-00-937-6156
Marconi	09553	893C	6625-01-282-9397
Avionics Specialists Inc.	8S585	ASI-83C	6625-01-441-9555

6.3 Source of Documents.

6.3.1 Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Defense Documents Order Desk, 700 Robins Ave., Bldg 4D, Philadelphia PA 19111-5094. Ph: (215 697-2179)

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